

Chapter 30: Baranaja: Multicropping Indigenous System of Uttarakhand Himalaya as a Model of Sustainable Agriculture and Food Security

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Abstract

Baranaja is a traditional multi-cropping and experience-based model developed by local communities in the mountain agriculture of the Uttarakhand Himalayas. This model has played an important role in the rural economy along with livelihood, food security, and nutrition of the local community for centuries. Moreover, this indigenous system has been instrumental in conserving local natural resources. The present chapter analyses this traditional system in the context of sustainable agriculture and food security. The study analysed the agro-ecological characteristics of Baranaja, its potential for crop diversification, biodiversity and water conservation, adaptation to climate change, etc. The socio-economic utility of the system from an economic perspective, its contribution to rural livelihoods, community participation, and nutrition and food security are also covered in the chapter. The chapter presents policy and institutional suggestions for the preservation and promotion of the system while analysing the challenges currently faced by it. Thus, this system, based on the Indian Himalayan knowledge tradition as a whole, presents significant possibilities towards sustainable mountain development.

Keywords: *Baranaja, Sustainable Agriculture, Food and Nutrition Safety, Traditional Agriculture, Multicropping System.*

1.0 Introduction

Agriculture is currently facing various challenges globally, including major ones such as climate change, biodiversity loss, soil degradation, and food nutritional security (Jamod et al., n.d.; Chand, 2022; Balkrishna et al., 2021). These challenges are even more severe in the case of agriculture in hilly terrains as factors such as the fragility of the ecosystem, year dependence of agriculture, small and fragmented holdings, and limited availability of external inputs act to increase the vulnerability of agriculture in these terrains (Pratap, 2011; Das et al., 2018; Sati et al., n.d.; Joshi & Dwivedi, 2025). In such a scenario, indigenous farming systems based on traditional Himalayan local knowledge of these regions are beginning to attract the attention of the present world.

The multicrop farming system prevalent in the Uttarakhand Himalayas, known as the Baranaja system in the local language, is one such indigenous farming system (Ghosh & Dhyani, 2004; Down to Earth, 2016). This locally developed farming system presents an effective model of sustainable agriculture and food security. For this reason, this system is considered an appropriate for the fragile condition of this hilly terrain (Gururani et al., 2021; Ghosh & Dhyani, 2004; Chowdhury et al., 2025).

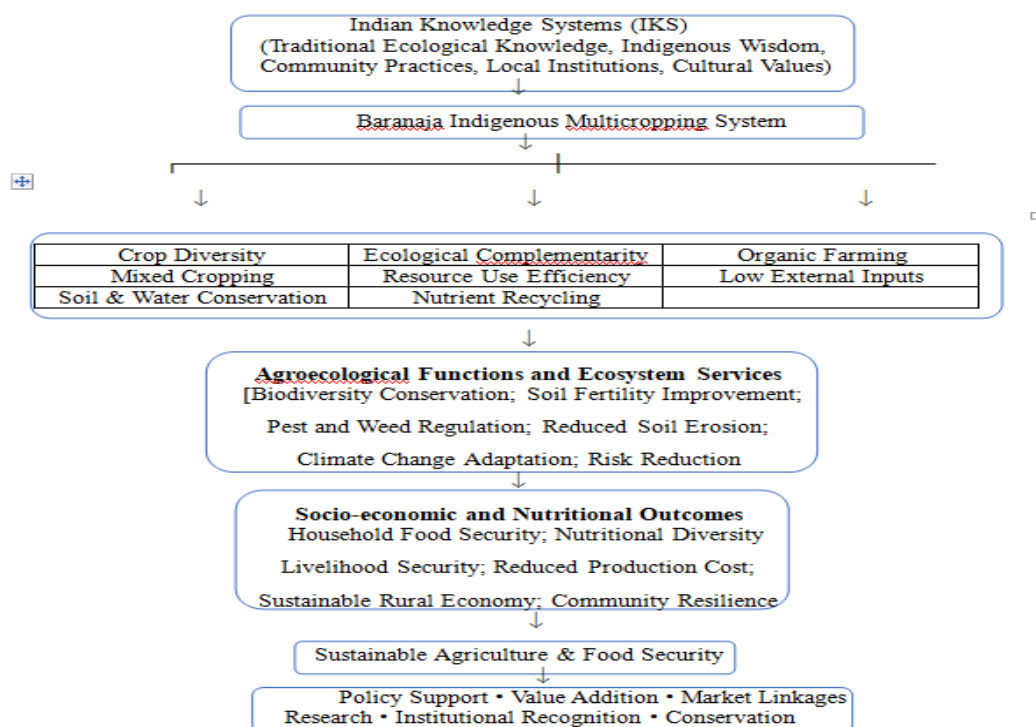
The literal meaning of Baranaja is twelve grains (Jardhari, 2007,2012; Kala, 2010). It has been grown for centuries by farmers in this mountainous terrain as a multifarious and multi-crop system. Under this system, crops such as millets, pulses, oilseeds, vegetables, spices, and fibre are grown together during the rainy season in a single terraced field in a hilly terrain (Pande & Pande, 2004; Kala, 2010; Ghosh & Dhyani, 2004). The basis of the stability of this system is the ecological complementarity that exists between crops. Not only is it of ecological importance, but this agricultural system also ensures domestic food and nutrition.

The present chapter analyses this Himalayan traditional multi-cropped indigenous farming system as a holistic model of sustainable agriculture and food security, while also highlighting the main challenges faced by the system and the policy and institutional measures required for its sustained revival in the Himalayan Division.

2.0 Conceptual Framework: Indian Knowledge Systems and Sustainable Agriculture

The conceptual framework of the present chapter covers four major dimensions: traditional agricultural systems and knowledge passed on from generation to generation, agroecological practices, ecological and social consequences, and institutional and policy support. Figure 1 presents this conceptual framework:

Figure 1: Conceptual Framework



3.0 Baranaja Farming System: An Overview

Baranaja is an ancient indigenous multi-cropped agricultural system based on Himalayan knowledge of Uttarakhand (Ghosh & Dhyani, 2004; Gururani et al., 2021). This system, consisting of a mix of crops such as cereals, pulses, oilseeds, etc., can vary from region to region depending on the number of crop types, climatic conditions, altitude, food traditions, and agricultural requirements (Jardhari, 2007,2012; Meena & Dawar, 2022). In practice, farmers sometimes cultivate more than twelve crops, although Jardhari (2007, 2012) describes these as varying by region, geography, and diet. The composition of these crops varies, influenced by altitude, local agroecological conditions, family food requirements, and traditional agricultural knowledge. This is why this cropping system can be understood as a flexible and site-specific agricultural system based on spatial knowledge and ecology, rather than as a fixed crop combination (Gururani et al., 2021; Kala, 2010).

The basic premise of this system is cropping diversity as well as ecological complementarity, in which diverse crops are selected in such a way that they mutually meet each other's nutritional, physical, and ecological requirements. This can be understood as, for example, leguminous crops are helpful in increasing soil fertility as well as in stabilising atmospheric nitrogen, so crops that have deep and strong roots prevent soil erosion, help other upright crops to grow upwards as support for creeper crops, provide fodder for livestock, etc., are mutually helpful. The effect is that the system maximises the use of natural resources, minimises production risk, and ensures the sustainability of the agricultural system (Maikhuri et al., 1997; Gururani et al., 2021; Lohani et al., 2025).

Baranaja cropping system consists of a diverse combination of seeds and it varies across villages and agroecological conditions, but each crop performs a distinct ecological, nutritional and economic function within the farming system (Chamoli et al., 2024; Jarhdari, 2007-2012; ithy.com; Meena & Dawar, 2022; Ghosh & Dhyani, 2004).

Table 1. Some of the Major Crop Groups in the Baranaja Farming System and Their Functional Roles

Crop Group	Major Crops	Primary Function	Additional Benefit
Millets	Mandua (<i>Eleusine coracana</i>), Jhangora (<i>Echinochloa frumentaceae</i>), Ramdana (<i>Amaranthus frumentaceus</i>), Ugal (<i>Fagopyrum esculentum</i>), Makka (<i>Zea mays</i>)	Staple food	Climate resilient, rich in calcium
Pulses	Gahat (<i>Macrotyloma uniflorum</i>), Kala Bhatt (<i>Glycine spp.</i>), Rajma (<i>Phaseolus vulgaris</i>), Lobiya (<i>Vigna unguiculata</i>), Naurangi (), Urad (<i>Vigna mungo</i>), Rains (), Soonth (<i>Vigna unguiculata</i>), Moond (<i>Vigna radiata</i>)	Protein source	Nitrogen fixation
Oilseeds	Til (<i>Sesamum indicum</i>), Bhangjeera (<i>Perilla frutescens</i>), Bhanjeer (<i>Perilla frutescens</i>)	Edible oil	Healthy fats
Vegetables	Pumpkin (<i>Cucurbita moschata</i>), Cucumber (<i>Cucumis</i>)	Vitamins	Dietary

	<i>melo</i>), Gourd (<i>Lagenaria siceraria</i>) etc.		diversity
Spices	Jakhiya (<i>Cleome viscosa</i>), Turmeric (<i>Curcuma longa</i>)	Taste & medicine	Market value
Fodder/ Fibre	Local grasses	Livestock support	Circular farming

Source: Prepared by the author based on a synthesis of published literature

This traditional farming system, based on low inputs, relies on cow dung manure and other locally available organic resources rather than modern chemical pesticides and fertilisers. These have far-reaching consequences for soil health, biodiversity conservation, and long-term ecological balance, as well as relatively low agricultural costs (Meena & Dawar, 2022).

4.0 Baranaja and Sustainable Agriculture

This traditional multi-cropping system of the Uttarakhand Himalayas is considered an excellent indigenous model of sustainable agriculture as the system incorporates key elements such as conservation of natural resources, biodiversity conservation, and long-term balance of agroecosystems. The system is based on the combined cultivation of diverse crops as opposed to the modern monoculture system, enabling more efficient use of land, water, and nutrients. The result is visible in the form of stability in agricultural production (Ghosh & Dhyani, 2004; Gururani et al., 2021).

The crop diversity of this system makes it one of the most important agricultural systems. Under this, the production of diverse grains makes the agroecosystem more stable and balanced, while pulse crops increase the fertility level of the soil, and roots with different depths are able to better utilise nutrients and moisture as well as prevent soil erosion. This system thus helps in maintaining agricultural productivity while conserving natural resources (Maikhuri et al., 1997; Meena & Dawar, 2022).

This cropping system, being based mostly on non-chemical and local resources with low input costs, is considered to be able to preserve soil bio-quality, microbial activity, and environmental balance, as well as lower production costs (Meena & Dawar, 2022). The system also plays an important role in reducing agricultural risk and maintaining production stability. When a crisis occurs in one crop, other crops provide the basis of production and income to the farmer. This feature makes the perennial farming system a flexible and sustainable farming system for the hilly region (Gururani et al., 2021).

Combined cultivation of crops with characteristics such as variety of root structure, nutrient requirement, and maturity period serves to reduce the risk of production along with more efficient use of natural resources. This diversity thus leads to increased soil fertility through organic nutrient recycling, reduced soil erosion in terraced fields, controlled or reduced weed and pest infestation in crops, as well as reduced dependence on the use of chemical means. The resulting multi-crop system serves to strengthen agricultural adaptability to climate change and climatic uncertainties and contributes significantly to the long-term sustainability of the agriculture of the mountainous Himalayan terrain (Ghosh & Dhyani, 2004; Kala, 2010; Gururani et al., 2021). This mixed cropping system, therefore, enhances crop diversity as well as the resilience of agroecosystems and serves as an effective strategy to deal with climate and economic uncertainties (Bisht, 2020; Rana & Bisht, 2023; Negi & Maikhuri, 2013).



Figure 2: Contribution of Baranaja to Sustainable Agriculture

Table 2: Contribution of Baranaja to Sustainable Agriculture

Dimension	Contribution
Environmental Sustainability	Conserves biodiversity and natural resources
Soil Sustainability	Improves soil fertility through nutrient recycling
Water Sustainability	Efficient utilization of rainwater and soil moisture
Economic Sustainability	Reduces production cost and production risk
Social Sustainability	Strengthens community participation and local institutions
Climate Sustainability	Enhances resilience against climatic variability

Source: Prepared by the author based on a synthesis of published literature

5.0 Baranaja and Food Security

The under-appreciation of the long-lasting value of indigenous knowledge systems in helping people prepare for and lessen the effects of droughts and food insecurity (Kumabhar & Kharwar, 2023). This system is an excellent agricultural as well as a traditional model of food and nutrition security that has evolved based on the local needs, resources, and experiences of the hill communities over time. It ensures the availability of diverse and balanced diets to families; hence, this agricultural system can be considered a symbol of the concept of "Balanced farm, Balanced food and Balanced society."

The biggest feature of this system is cropping diversification; whereby different types of food grains are obtained from the same field from time to time. It provides a balanced food basket to the farmer through the production of cereals, vegetables, pulses, oilseeds and spices, etc., which maintains the year-round dietary diversity of local families. These traditional crops are widely recognised for their high nutritional value and special place in Himalayan food culture (Bisht et al., 2018; Ojha et al., 2022; Sharma, 2025). These traditional food systems are not limited to food availability but also contribute significantly to improving the nutritional quality of the family, taking care of community health and preserving this important traditional knowledge that is associated with consumption and production (Ojha et al., 2022).

Nutritionally, these crops are rich in protein, fibre, calcium, iron, zinc and other micronutrients. The availability of these nutrients contributes significantly to the prevention of malnutrition and better health through a balanced family diet (Gururani et al., 2021; Meena & Dawar, 2022; Chamoli et al., 2024). It is a system of risk distribution and continuous assurance of food availability, with a new possibility of food crisis if one crop is affected by some kind of famine or crisis, but the production of other crops is achieved (Lohani et al., 2025).

From an economic point of view, this cropping system is a strong base of food security. This system reduces their dependence on the external market and ensures that the family's food needs are met locally by hill farmers who operate their agriculture on limited land and resources.

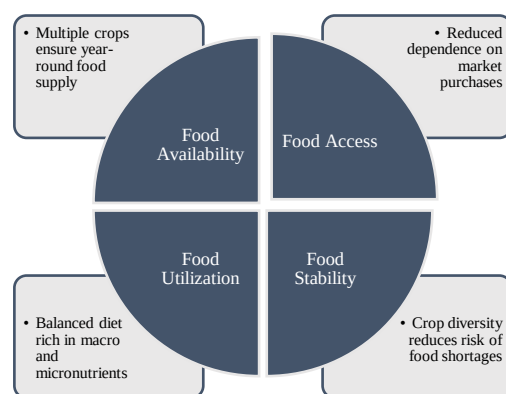


Figure 3: Contribution of Baranaja to the Four Dimensions of Food Security*

Source: Prepared by the author based on a synthesis of published literature (*Four dimensions of food security defined by Food and Agricultural Organisation (FAO))

6.0 Socio-Economic Dimensions of Baranaja System

Baranaja is not only a traditional agricultural system of Uttarakhand but is also embedded in the social, cultural and economic life system of the hill society. It has the potential to play an important role in improving the rural economy as it incorporates traditional Himalayan knowledge and local resource-based elements in a spirit of community cooperation. Traditionally, the inclusion of collective labour in this practice enhances the sense of social cohesion and mutual support, or even replaces economic capital with social capital, as agricultural operations such as sowing, weeding, harvesting, threshing, or composting require more human labour.

From an economic point of view, Baranaja is a low-cost as well as low-risk farming method for hill farmers comparatively. In this method, the cost of production is controlled, and crop residues supplement the livestock feed. In such a situation, it can be seen as a strong integrated system between agriculture and animal husbandry, as farmers in hilly areas practice animal husbandry to support their livelihood along with agriculture. The increasing market demand for some coarse grains and oilseeds paves the way for strengthening rural livelihoods by generating additional income from the surplus production of farmers.

The association of this cropping system with the local culture also explains its cultural significance, as the various crops grown here are an important part of the local hill and religious rituals and traditional

cuisine. If we look at this system from the perspective of small and marginal farmers in particular, it has been an effective means of protecting their livelihood. This system not only strengthens livelihoods and food security but also helps local culture contribute significantly to community relations and conservation of agro-biodiversity.

Table 3. Socio-economic Contributions of the Baranaja System

Dimension	Contribution	Long-term Outcome
Household Food Security	Diverse food basket	Improved nutrition
Rural Livelihood	Multiple farm outputs	Stable income
Livestock Integration	Fodder availability	Sustainable mixed farming
Community Cooperation	Collective labour and knowledge sharing	Social cohesion, replacement of economic capital through social capital
Local Economy	Sale of surplus produce	Rural economic development

Source: Prepared by the author based on a synthesis of published literature

7.0 Challenges and Decline of Baranaja System

Presently, the perennial system is on the verge of marginalisation due to various challenges in hill agriculture. Some of the major challenges it faces can be summarised as follows:

- **Migration** - Continuous migration of youth from hilly terrains due to various reasons has created a crisis for agriculture.
- **Loss of traditional knowledge** - The effect of the gradual loss of traditional indigenous knowledge due to the lack of generational transfer is seen as a challenge to the reforestation system, as agricultural management, seed conservation, and traditional indigenous agricultural knowledge are lost.
- **Preference to single cropping system**- In the present scenario, agro-commercialisation has increased the impact of the single cropping system. As a result of shifting to a mono-cropping system and neglecting multi-cropping, challenges are being posed to agro-biodiversity as well as ecological balance.
- **Challenges to agriculture from wild animals**- Challenges such as destruction of crops by wild animals pose a threat to terrestrial agriculture, as farmers are abandoning it and adopting other alternative means of livelihood as agriculture becomes riskier.
- **Limited market and single market**- The lack of an organised market for the agricultural produce of the hilly region, the challenges of value addition and an effective marketing system affect and limit the income of the farmers, which also affects agricultural production.

- **Policy and Institutional Support-** While the acceptance of an agricultural system based on indigenous knowledge has increased, the support of symptomatic policies and institutions is still needed for its proper preservation and expansion of the Baranaja agriculture system.

8.0 Policy Implications and Way Forward

Baranaja is a traditionally hill farming system in the hills of Uttarakhand, that offers significant potential for rural development and sustainable agriculture. In such a situation, its revival in the present context can give a multi-faceted effect. It will need to be given policy recognition as a viable agricultural system, beyond mere cultural or traditional heritage.

- This multifaceted agriculture will need to be provided with appropriate promotion and protection (agricultural incentives, indigenous seed conservation programmes, financial inclusion, and market opportunities) at the central or state government level.
- It will be necessary to promote research related to this system in agricultural universities and research institutes to make it effective and farmer-friendly according to the current times, so that this traditional system can be developed on a sound scientific basis and made more beneficial to the farmer.
- Under direct contact and interaction with the farmers, a proper training program and timing will be provided through Krishi Vigyan Kendras (KVK), the Department of Agriculture, the Department of Fruit Horticulture and other agriculture-related training institutes.
- Direct contact with the local rural community through self-help groups, at the NGO level, can help in taking the campaign forward. The fear of its extinction can be controlled by conducting programs such as 'indigenous seed conservation programs' from time to time, documenting traditional agricultural knowledge and transferring the knowledge of experienced farmers through the group to future generations.
- Linking local women's self-help groups to the processing and entrepreneurship of agri products for women's empowerment can help increase women's income-earning as well as their economic empowerment, as women have a special role in agriculture in hilly terrains.
- It will be necessary to connect farmer development organisations and agricultural cooperatives with it and make the products accessible to the organised market through them for better market availability of agricultural products. Digital marketing of products, organic certification, GI tags, quality standardisation, etc. can help structure a larger market for these products that can have access to the international market.

Hence, these effective steps can go a long way in establishing the perennial farming system as a robust, resilient and sustainable agricultural system in the hilly terrain of the state.

9.0 Conclusion

The traditional multi-cropped ecosystem of the Uttarakhand hill tracts is established as not just an agricultural system but a larger agricultural model based on the Indian Himalayan knowledge tradition that integrates agro-biodiversity conservation, appropriate use of natural resources, climate adaptation, as well as rural livelihood sustainability and food nutrition security. This system has been popular with local communities for centuries as a self-sufficient agricultural system and risk tolerance, possible in many situations such as sloping small or marginal agricultural land with heterogeneous hilly topography,

dependent on rainfall, low external inputs, etc. Traditionally important for strengthening the local rural economy, the system is currently struggling for survival in the face of various challenges.

Thus, this Himalayan traditional system proves that the Indian knowledge tradition is not only a cultural heritage but also helpful in presenting solutions to contemporary agricultural crises, and the basis of these solutions is practical and scientific. At present, when the agrarian crisis is intensifying, and this multifaceted traditional system of the hilly region is emerging as a viable scientific and sustainable agricultural model, it will be necessary to patronise this traditional and adaptive system. If a very holistic approach is adopted for its conservation and promotion, it can emerge as an exemplary model for the development of the entire hill agricultural regions.

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